

$$\begin{aligned}\ell(A, \alpha) &= (N_T - 1) \log(A\alpha) - (1 + \alpha) \sum_{n=1}^{N_T-1} \log(Ax_{l_{n+1}}) - \sum_{n=1}^{N_T-1} (Ax_{l_{n+1}})^{-\alpha} \\ &\quad - \sum_{n=1}^{N_T-1} (\Delta_{n+1} - 1)(Ax_{l_n})^{-\alpha} - (T - l_{N_T})(Ax_{l_{N_T}})^{-\alpha}\end{aligned}$$

For compactness, define the index sets:

$$S_1 = \{x_{l_{n+1}}\}_{n=1}^{N_T-1}, \quad S_2 = \{x_{l_n}\}_{n=1}^{N_T-1}, \quad x_* = x_{l_{N_T}}.$$

1. First derivatives

Derivative w.r.t. A

Recall:

$$\frac{\partial}{\partial A} \log(A) = \frac{1}{A}, \quad \frac{\partial}{\partial A} (Ax)^{-\alpha} = -\alpha x (Ax)^{-\alpha-1}.$$

Result:

$$\begin{aligned}\frac{\partial \ell}{\partial A} &= \frac{N_T - 1}{A} - (1 + \alpha) \frac{N_T - 1}{A} \\ &\quad + \alpha \sum_{n=1}^{N_T-1} x_{l_{n+1}} (Ax_{l_{n+1}})^{-\alpha-1} \\ &\quad + \alpha \sum_{n=1}^{N_T-1} (\Delta_{n+1} - 1) x_{l_n} (Ax_{l_n})^{-\alpha-1} \\ &\quad + \alpha (T - l_{N_T}) x_{l_{N_T}} (Ax_{l_{N_T}})^{-\alpha-1}\end{aligned}$$

You can simplify the first line:

$$\frac{N_T - 1}{A} - (1 + \alpha) \frac{N_T - 1}{A} = -\frac{\alpha(N_T - 1)}{A}.$$

Derivative w.r.t. α

Recall:

$$\frac{\partial}{\partial \alpha} (Ax)^{-\alpha} = -(Ax)^{-\alpha} \log(Ax).$$

Result:

$$\begin{aligned} \frac{\partial \ell}{\partial \alpha} = & \frac{N_T - 1}{\alpha} - \sum_{n=1}^{N_T-1} \log(Ax_{l_{n+1}}) \\ & + \sum_{n=1}^{N_T-1} (Ax_{l_{n+1}})^{-\alpha} \log(Ax_{l_{n+1}}) \\ & + \sum_{n=1}^{N_T-1} (\Delta_{n+1} - 1)(Ax_{l_n})^{-\alpha} \log(Ax_{l_n}) \\ & + (T - l_{N_T})(Ax_{l_{N_T}})^{-\alpha} \log(Ax_{l_{N_T}}) \end{aligned}$$

2. Second derivatives

Second derivative w.r.t. A

Using:

$$\frac{\partial^2}{\partial A^2} (Ax)^{-\alpha} = \alpha(\alpha + 1)x^2(Ax)^{-\alpha-2}$$

Result:

$$\begin{aligned} \frac{\partial^2 \ell}{\partial A^2} = & \frac{\alpha(N_T - 1)}{A^2} \\ & - \alpha(\alpha + 1) \sum_{n=1}^{N_T-1} x_{l_{n+1}}^2 (Ax_{l_{n+1}})^{-\alpha-2} \\ & - \alpha(\alpha + 1) \sum_{n=1}^{N_T-1} (\Delta_{n+1} - 1) x_{l_n}^2 (Ax_{l_n})^{-\alpha-2} \\ & - \alpha(\alpha + 1) (T - l_{N_T}) x_{l_{N_T}}^2 (Ax_{l_{N_T}})^{-\alpha-2} \end{aligned}$$

Second derivative w.r.t. α

Using:

$$\frac{\partial}{\partial \alpha} [(Ax)^{-\alpha} \log(Ax)] = -(Ax)^{-\alpha} (\log(Ax))^2$$

Result:

$$\begin{aligned} \frac{\partial^2 \ell}{\partial \alpha^2} = & -\frac{N_T - 1}{\alpha^2} \\ & - \sum_{n=1}^{N_T-1} (Ax_{l_{n+1}})^{-\alpha} (\log(Ax_{l_{n+1}}))^2 \\ & - \sum_{n=1}^{N_T-1} (\Delta_{n+1} - 1) (Ax_{l_n})^{-\alpha} (\log(Ax_{l_n}))^2 \\ & - (T - l_{N_T}) (Ax_{l_{N_T}})^{-\alpha} (\log(Ax_{l_{N_T}}))^2 \end{aligned}$$

Cross derivative $\frac{\partial^2 \ell}{\partial A \partial \alpha}$

Using product rule:

$$\frac{\partial}{\partial A} [(Ax)^{-\alpha} \log(Ax)] = -\alpha x (Ax)^{-\alpha-1} \log(Ax) + \frac{x}{A} (Ax)^{-\alpha}$$

Result:

$$\begin{aligned} \frac{\partial^2 \ell}{\partial A \partial \alpha} = & -\frac{N_T - 1}{A} \\ & + \sum_{n=1}^{N_T-1} x_{l_{n+1}} (Ax_{l_{n+1}})^{-\alpha-1} [1 - \alpha \log(Ax_{l_{n+1}})] \\ & + \sum_{n=1}^{N_T-1} (\Delta_{n+1} - 1) x_{l_n} (Ax_{l_n})^{-\alpha-1} [1 - \alpha \log(Ax_{l_n})] \\ & + (T - l_{N_T}) x_{l_{N_T}} (Ax_{l_{N_T}})^{-\alpha-1} [1 - \alpha \log(Ax_{l_{N_T}})] \end{aligned}$$